

Dataset Overview

This dataset provides insights into personal finance and investment preferences across a diverse group of individuals. It includes demographic information, preferred investment avenues, decision-making factors, financial goals, and sources of financial information.

Each record represents the investment profile of a single respondent.

Column Descriptions

- **Gender / Age**
 - Represents the respondent's gender and age.
- **Investment_Avenues**

Indicates whether the respondent actively invests (Yes/No).

- **Mutual_Funds, Equity_Market, Debentures, Government_Bonds, Fixed_Deposits, PPF, Gold**

Ranked preference scores (1–7) for each investment type, reflecting their relative priority.

- **Stock_Market**

Indicates whether the respondent participates in the stock market (Yes/No).

- **Factor**

The primary factor influencing investment decisions (e.g., Returns, Risk).

- **Objective**

The main investment objective (e.g., Capital Appreciation, Growth, Income).

- **Purpose**

The broader financial goal (e.g., Wealth Creation, Future Savings, Returns).

- **Duration**

Preferred investment time horizon (e.g., 3–6 months, 1–3 years, 5–10 years).

- **Invest_Monitor**

Frequency of monitoring investments (Daily, Weekly, Monthly).

- **Expect**

Expected return range (e.g., 10%–20%, 20%–30%, 30%–40%).

- **Avenue**

Most preferred investment avenue (e.g., Mutual Funds, Equity, Fixed Deposits, PPF).

- **Savings Objectives**

Specific financial goals (e.g., Retirement Planning, Education, Healthcare).

- **Reason_Equity, Reason_Mutual, Reason_Bonds, Reason_FD**

Key reasons for selecting each investment type (e.g., Capital Appreciation, Tax Benefits, Safety, Fixed Returns).

- **Source**

Primary source of financial information (e.g., Newspapers, Television, Internet, Financial Consultants).

Tools Used and Components

A) Alteryx Components

1. Input Data Tool

- Used to load the raw dataset from sources such as Excel and OLE DB (SQL Server).

2. Select Tool

- Used to:
 - Select required fields and remove unnecessary ones

- Rename columns
- Modify data types

3. Formula Tool

- Used to create new calculated fields and apply transformations.

4. Filter Tool

- Used to split data based on specific conditions.

5. Record ID Tool

- Used to generate a unique identifier for each record.

6. Summarize Tool

- Used to aggregate data (e.g., sum, count, average).

7. Append Fields Tool

- Used to combine data from different sources.

8. Text to Columns Tool

- Used to split fields into multiple columns based on a delimiter.

9. Join Tool

- Used to merge datasets based on common fields.

10. Output Data Tool

- Used to export the processed data to a destination (e.g., Excel, SQL Server).

1. Dataset Splitting (Dividing Records into Two Halves)

The screenshot displays the Alteryx Designer interface for a workflow named 'split.ymd'. The workflow starts with an input tool 'Finance_data.csv'. It then passes through a 'Select' tool, a 'Sort' tool, and a 'Join' tool. The output of the 'Join' tool is connected to a 'Split' tool, which is configured to split the data into two equal halves based on the 'RecordID' field. The two resulting streams are then loaded into two separate Excel files: 'First_half.xlsx Query-Sheet1' and 'Second_half.xlsx Query-Sheet1'. The left sidebar shows the 'Output Data - Configuration' panel, which is set to 'Write to File or Database' with the path 'C:\Users\engma\Downloads\First_half.xlsx\Sheet1'. The bottom panel shows the 'Results - Output Data (6) - Input' section, which is currently empty, displaying the message 'No data available. Use Ctrl+R to run the workflow.'

2. Data Transformation Using Alteryx

The screenshot displays the Alteryx Designer interface for a workflow named 'final.ymd'. The workflow starts with an input tool 'First_half.xlsx Query-Sheet1'. It then passes through a 'Select' tool, a 'Sort' tool, and a 'Join' tool. The output of the 'Join' tool is connected to a 'Split' tool, which is configured to split the data into two equal halves based on the 'RecordID' field. The two resulting streams are then loaded into two separate Excel files: 'First_half.xlsx Query-Sheet1' and 'Second_half.xlsx Query-Sheet1'. The left sidebar shows the 'Workflow - Configuration' panel, which is set to 'Canvas' with the layout direction 'Horizontal'. The bottom panel shows the 'Results - Workflow - Messages' section, which is currently empty, displaying the message 'No data available. Use Ctrl+R to run the workflow.'

3. Loading Dimension Tables

The image displays two screenshots of the Alteryx Designer x64 interface, illustrating the process of loading dimension tables into a data warehouse.

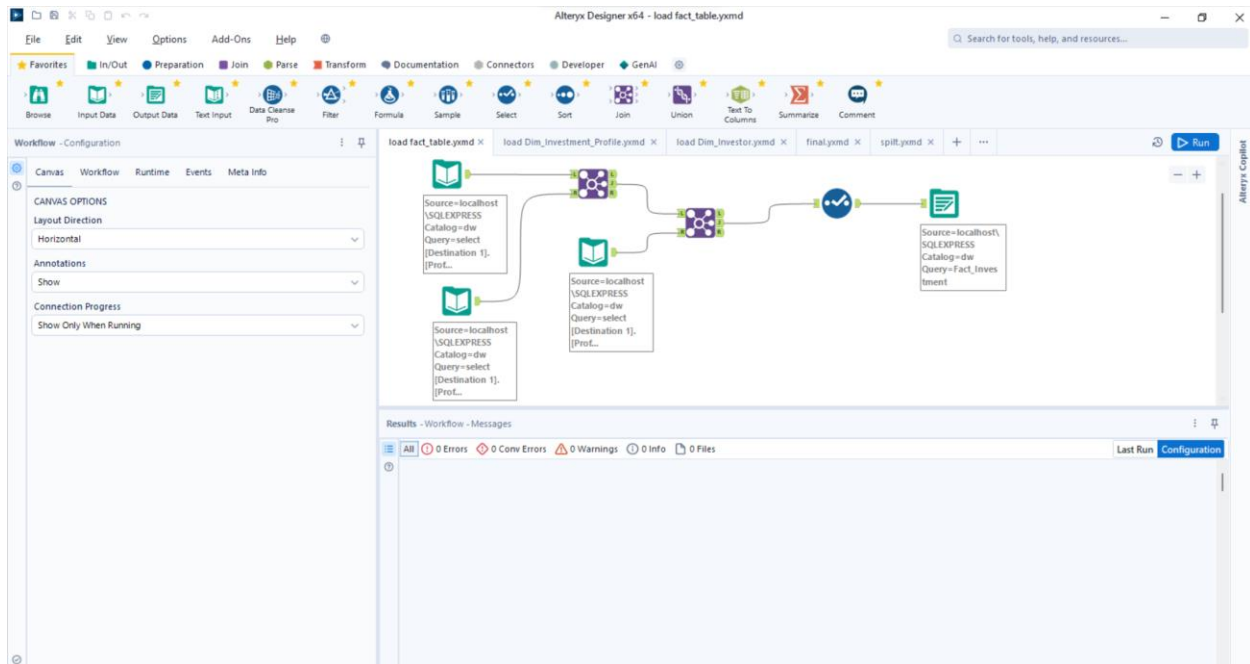
Top Screenshot: load Dim_Investment_Profile.yxdm

- Canvas:** Shows a workflow with three main components: an Input Data tool (Source=localhost\SQL EXPRESS Catalog=dw Query=select [Destination 1]. [Prof...]), a Join tool, and an Output Data tool (Source=localhost\SQL EXPRESS Catalog=dw Query=Dim_Inves tment_Profile).
- Workflow - Configuration:** The left sidebar shows the configuration for the workflow. Under "CANVAS OPTIONS", the "Layout Direction" is set to "Horizontal".
- Results - Workflow - Messages:** The bottom panel shows the results of the workflow, indicating 0 Errors, 0 Conv Errors, 0 Warnings, 0 Info, and 0 Files.

Bottom Screenshot: load Dim_Investor.yxdm

- Canvas:** Shows a similar workflow structure to the top screenshot, with an Input Data tool, a Join tool, and an Output Data tool (Source=localhost\SQL EXPRESS Catalog=dw Query=Dim_Inves tor).
- Workflow - Configuration:** The left sidebar shows the configuration for the workflow. Under "CANVAS OPTIONS", the "Layout Direction" is set to "Horizontal".
- Results - Workflow - Messages:** The bottom panel shows the results of the workflow, indicating 0 Errors, 0 Conv Errors, 0 Warnings, 0 Info, and 0 Files.

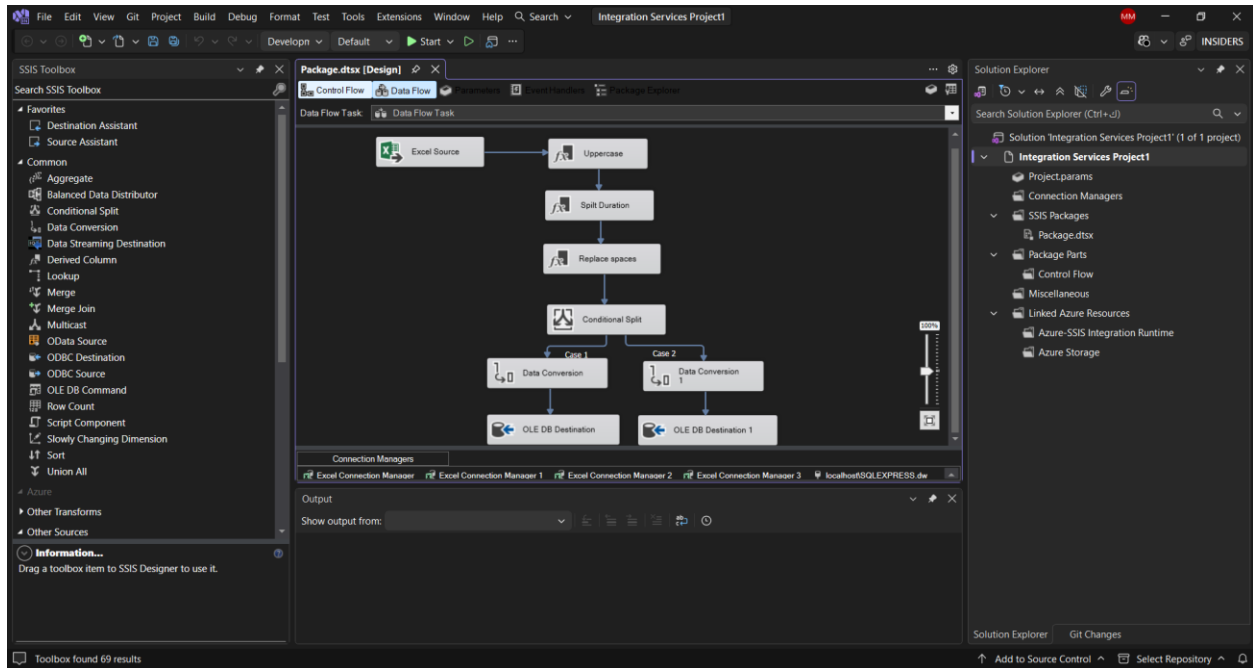
4. Loading the Fact Table



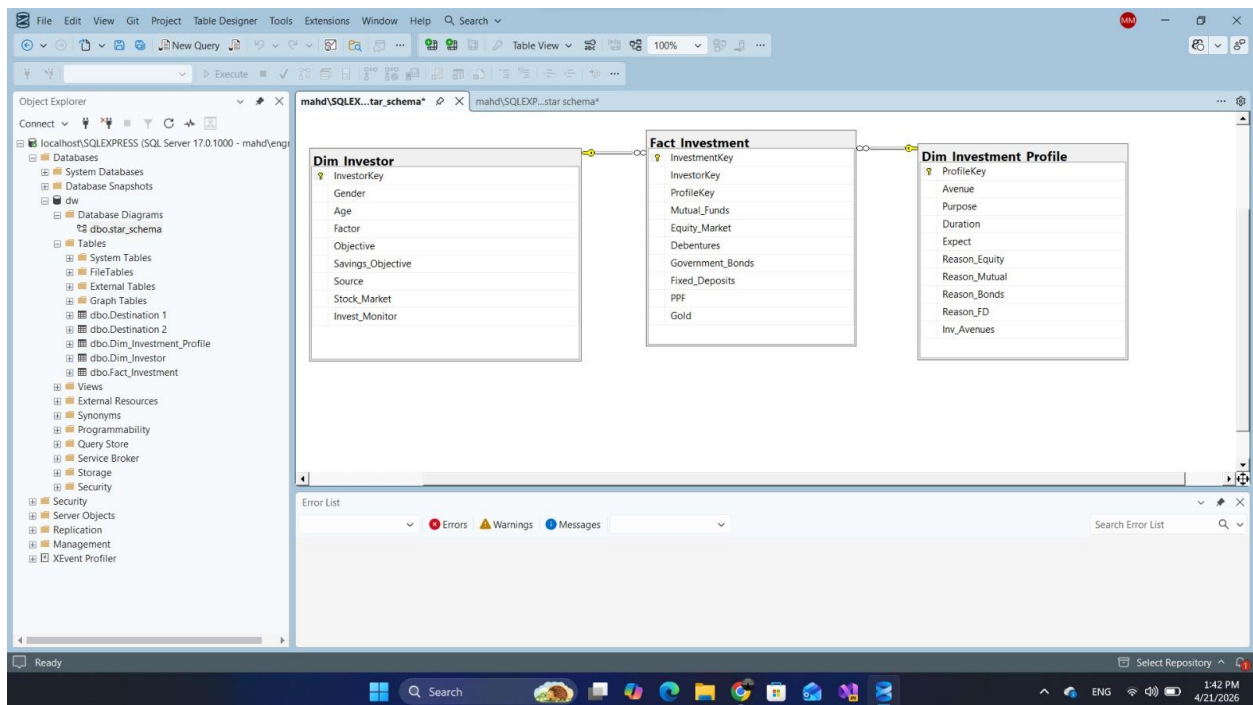
B) SSIS Components

- **Data Flow Task**
- **Excel Source**
- **Conditional Split**
- **Derived Column**
- **OLE DB Destination**

workflow



Star Schema



DESTINATION 1

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'localhost\SQLEXPRESS (SQL Server 17.0.1000 - mahd/engma)'. The central pane shows a query window with the following SQL code:

```
SELECT TOP (1000) [ProfileKey]
,[InvestorKey]
,[gender]
,[age]
,[Investment_Avenues]
,[Mutual_Funds]
,[Equity_Market]
,[Debtentures]
,[Government_Bonds]
,[Fixed_Deposits]
,[PPF]
,[Gold]
```

The query results are displayed in a table with the following columns: ProfileKey, InvestorKey, gender, age, Investment_Avenues, Mutual_Funds, Equity_Market, Debtentures, Government_Bonds, Fixed_Deposits, PPF, Gold, Stock_Market, Factor, Objective, Purpose, and Duration. The table shows 10 rows of data.

ProfileKey	InvestorKey	gender	age	Investment_Avenues	Mutual_Funds	Equity_Market	Debtentures	Government_Bonds	Fixed_Deposits	PPF	Gold	Stock_Market	Factor	Objective	Purpose	Duration
1	900	Male	27	Yes	2	3	6	5	4	1	7	Yes	Returns	Capital Appreciation	Wealth Creation	2-4 weeks
2	902	Male	30	Yes	3	6	4	2	5	1	7	Yes	Returns	Capital Appreciation	Wealth Creation	3-6 months
3	904	Male	27	Yes	2	4	7	5	1	3	6	Yes	Returns	Capital Appreciation	Wealth Creation	4-8 weeks
4	905	Male	26	Yes	2	3	6	4	1	5	7	Yes	Returns	Capital Appreciation	Wealth Creation	5-10 years
5	906	Male	26	Yes	2	3	6	4	1	5	7	Yes	Returns	Capital Appreciation	Wealth Creation	5-10 years
6	907	Male	27	Yes	3	2	7	4	5	1	6	Yes	Returns	Capital Appreciation	Wealth Creation	6-12 months
7	908	Male	29	Yes	4	3	5	7	2	1	6	Yes	Returns	Capital Appreciation	Wealth Creation	5-10 years
8	910	Male	29	Yes	2	3	6	5	1	4	7	Yes	Returns	Capital Appreciation	Wealth Creation	12-18 month
9	912	Male	29	Yes	2	3	6	5	1	4	7	Yes	Returns	Capital Appreciation	Wealth Creation	12-18 month

DESTINATION 2

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'localhost\SQLEXPRESS (SQL Server 17.0.1000 - mahd/engma)'. The central pane shows a query window with the following SQL code:

```
SELECT TOP (1000) [ProfileKey]
,[InvestorKey]
,[gender]
,[age]
,[Investment_Avenues]
,[Mutual_Funds]
,[Equity_Market]
,[Debtentures]
,[Government_Bonds]
,[Fixed_Deposits]
,[PPF]
,[Gold]
```

The query results are displayed in a table with the following columns: ProfileKey, InvestorKey, gender, age, Investment_Avenues, Mutual_Funds, Equity_Market, Debtentures, Government_Bonds, Fixed_Deposits, PPF, Gold, Stock_Market, Factor, Objective, Purpose, and Duration. The table shows 10 rows of data.

ProfileKey	InvestorKey	gender	age	Investment_Avenues	Mutual_Funds	Equity_Market	Debtentures	Government_Bonds	Fixed_Deposits	PPF	Gold	Stock_Market	Factor	Objective	Purpose	Duration
1	1	Female	34	Yes	1	2	5	3	7	6	4	Yes	Returns	Capital Appreciation	Wealth Creation	3-6 month
2	3	Female	24	No	7	5	4	6	3	1	2	No	Risk	Capital Appreciation	Wealth Creation	5-10 year
3	4	Female	27	Yes	3	6	4	2	5	1	7	Yes	Returns	Capital Appreciation	Wealth Creation	5-10 year
4	8	Female	35	Yes	2	4	7	5	3	1	6	Yes	Risk	Growth	Savings for Future	5-10 year
5	10	Female	21	No	1	2	3	4	5	6	7	No	Returns	Capital Appreciation	Wealth Creation	12-18 mo
6	11	Female	28	Yes	2	3	7	4	5	1	6	Yes	Returns	Capital Appreciation	Wealth Creation	3-6 month
7	12	Female	25	Yes	2	3	7	5	4	1	6	Yes	Returns	Capital Appreciation	Wealth Creation	3-5 years
8	14	Female	28	Yes	3	2	7	5	4	1	6	Yes	Risk	Growth	Wealth Creation	3-6 month
9	19	Female	24	Yes	2	4	5	6	3	1	7	Yes	Risk	Capital Appreciation	Wealth Creation	3-5 years

Dim Investment Profile

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure, including the 'dbo' schema and the 'Dim_Investment_Profile' table. The central pane shows a SQL query executed successfully, returning 616 rows. The query is as follows:

```
SELECT TOP (1000) [ProfileKey]
, [Avenue]
, [Purpose]
, [Duration]
, [Expect]
, [Reason_Equity]
, [Reason_Mutual]
, [Reason_Bonds]
, [Reason_FD]
, [Inv_Avenues]
FROM [dw].[dbo].[Dim_Investment_Profile]
```

The results pane displays the following data:

ProfileKey	Avenue	Purpose	Duration	Expect	Reason_Equity	Reason_Mutual	Reason_Bonds	Reason_FD	Inv_Avenues
1	Equity	Wealth Creation	3-6 months	20%-30%	Capital Appreciation	Tax Benefits	Assured Returns	Fixed Returns	Yes
2	Mutual fund	Wealth Creation	2-4 weeks	20%-30%	Capital Appreciation	Better Returns	Assured Returns	Risk Free	Yes
3	Equity	Savings for Future	3-6 months	20%-30%	Capital Appreciation	Fund Diversification	Safe Investment	Fixed Returns	Yes
4	Fixed deposits	Wealth Creation	1-3 years	30%-40%	Capital Appreciation	Fund Diversification	Assured Returns	Fixed Returns	Yes
5	Mutual fund	Wealth Creation	1-3 years	20%-30%	Capital Appreciation	Fund Diversification	Assured Returns	Fixed Returns	Yes
6	Mutual fund	Wealth Creation	2-4 weeks	20%-30%	Capital Appreciation	Fund Diversification	Assured Returns	Risk Free	Yes
7	Mutual fund	Wealth Creation	6-12 months	20%-30%	Capital Appreciation	Better Returns	Assured Returns	Risk Free	Yes
8	Mutual fund	Wealth Creation	1-3 years	20%-30%	Capital Appreciation	Better Returns	Assured Returns	Risk Free	Yes
9	Fixed deposits	Wealth Creation	4-5 weeks	20%-30%	Capital Appreciation	Fund Diversification	Assured Returns	Risk Free	Yes
10	Mutual fund	Wealth Creation	3-5 years	20%-30%	Capital Appreciation	Better Returns	Assured Returns	Fixed Returns	Yes

Dim Investor

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure, including the 'dbo' schema and the 'Dim_Investor' table. The central pane shows a SQL query executed successfully, returning 616 rows. The query is as follows:

```
SELECT TOP (1000) [InvestorKey]
, [Gender]
, [Age]
, [Factor]
, [Objective]
, [Savings_Objective]
, [Source]
, [Stock_Market]
, [Invest_Monitor]
FROM [dw].[dbo].[Dim_Investor]
```

The results pane displays the following data:

InvestorKey	Gender	Age	Factor	Objective	Savings_Objective	Source	Stock_Market	Invest_Monitor
1	Male	30	Returns	Capital Appreciation	Retirement Plan	Television	Yes	Daily
2	Male	21	Risk	Capital Appreciation	Retirement Plan	Newspapers_and_Magazines	Yes	Monthly
3	Male	35	Returns	Growth	Retirement Plan	Television	Yes	Weekly
4	Male	31	Returns	Capital Appreciation	Retirement Plan	Newspapers_and_Magazines	Yes	Monthly
5	Male	20	Risk	Capital Appreciation	Retirement Plan	Financial_Consultants	Yes	Monthly
6	Male	27	Returns	Capital Appreciation	Health_Care	Newspapers_and_Magazines	Yes	Monthly
7	Male	27	Returns	Capital Appreciation	Retirement Plan	Financial_Consultants	Yes	Monthly
8	Male	29	Risk	Capital Appreciation	Retirement Plan	Newspapers_and_Magazines	Yes	Monthly
9	Male	26	Risk	Capital Appreciation	Health_Care	Newspapers_and_Magazines	Yes	Monthly
10	Male	29	Returns	Growth	Retirement Plan	Financial_Consultants	Yes	Weekly

Fact Investment

The screenshot shows the Microsoft SQL Server Enterprise Manager interface. The left pane displays the Object Explorer with the database structure of 'localhost\SQLEXPRESS (SQL Server 17.0.1000 - mahd\engma)'. The central pane shows a SQL query in the 'SQLQuery5.sq...engma (75)' window. The query is a SELECT statement with TOP (1000) rows, listing various investment types and their counts. The bottom pane shows the results of the query, displaying a table with columns for InvestmentKey, InvestorKey, ProfileKey, Mutual_Funds, Equity_Market, Debentures, Government_Bonds, Fixed_Deposits, PPF, and Gold. The status bar indicates 'Query executed successfully.' and 'localhost\SQLEXPRESS (17.0... MAHD\engma (75) dw 00:00:00 Row: 1, Col: 1 616 rows'.

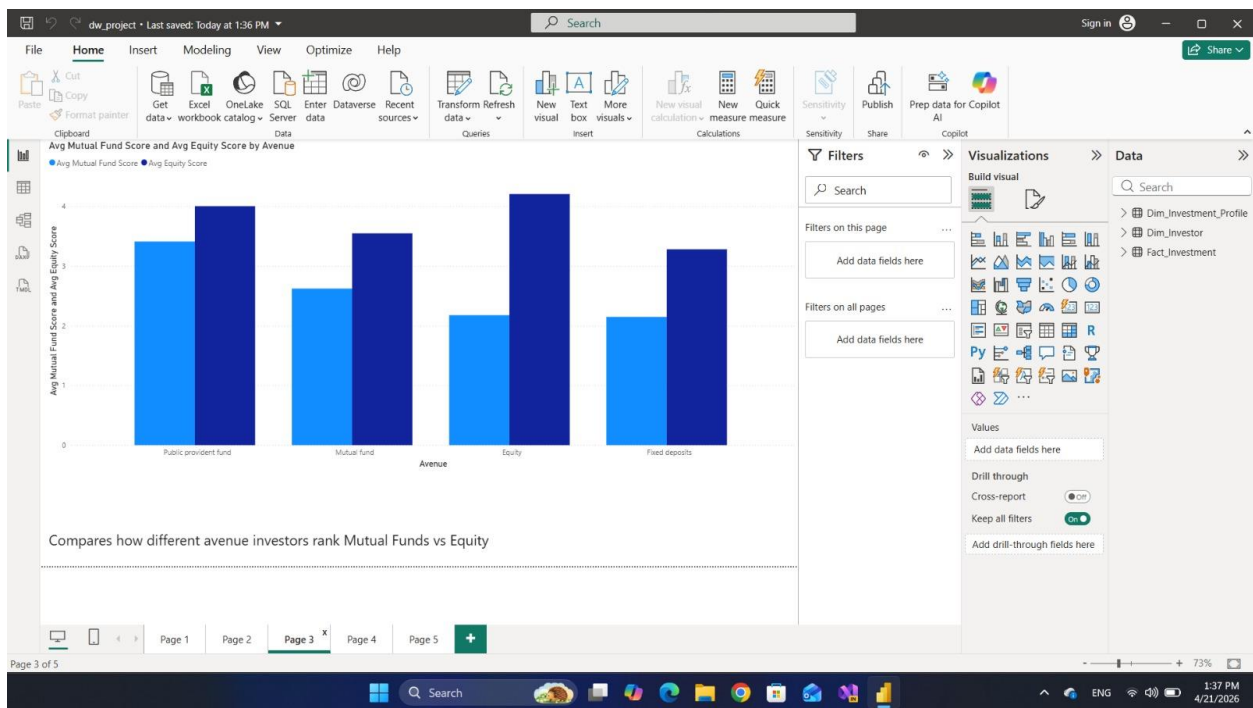
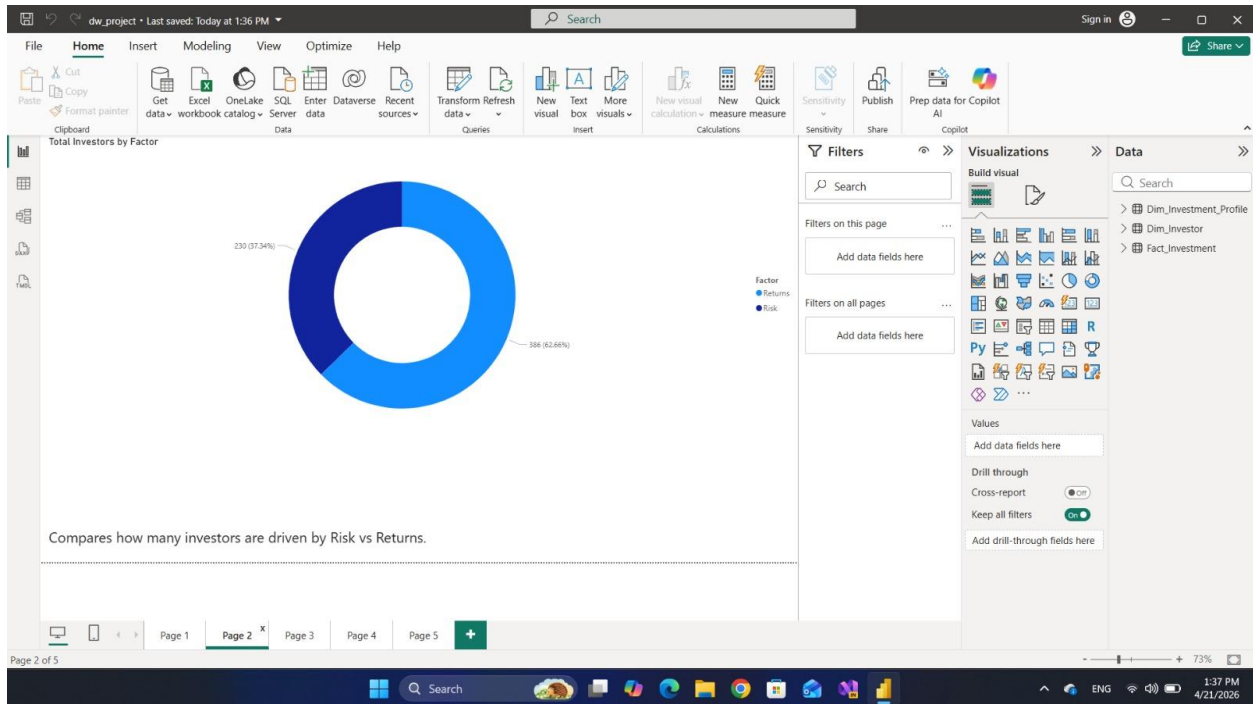
```
SELECT TOP (1000) [InvestmentKey]
,[InvestorKey]
,[ProfileKey]
,[Mutual_Funds]
,[Equity_Market]
,[Debentures]
,[Government_Bonds]
,[Fixed_Deposits]
,[PPF]
,[Gold]
FROM [dw].[dbo].[Fact_Investment]
```

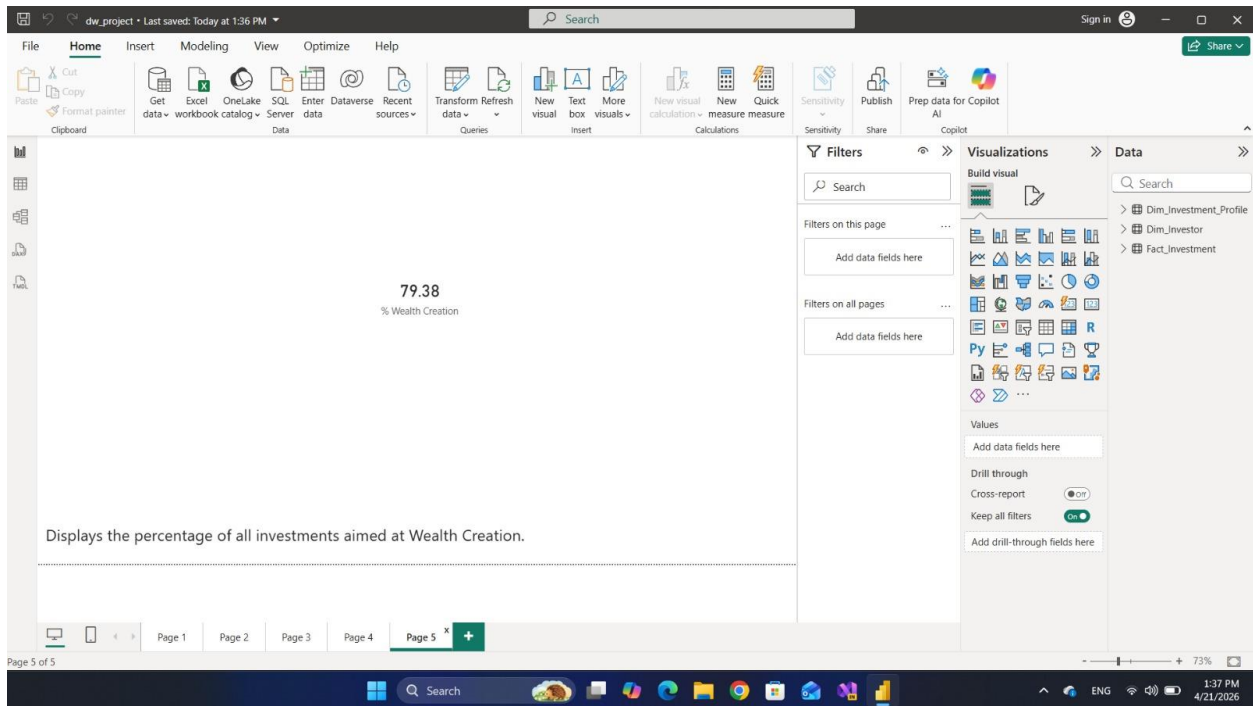
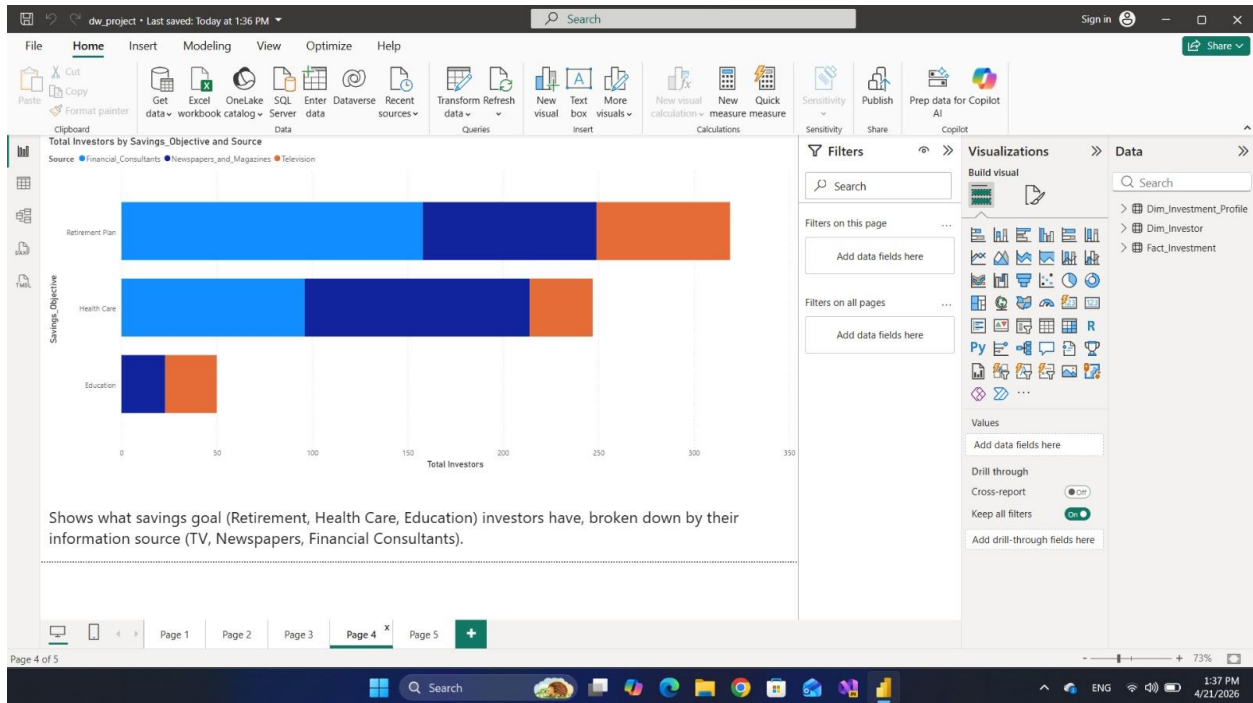
InvestmentKey	InvestorKey	ProfileKey	Mutual_Funds	Equity_Market	Debentures	Government_Bonds	Fixed_Deposits	PPF	Gold
1	1	543	4	5	1	2	7	3	6
2	2	762	4	3	5	7	2	1	6
3	3	872	2	4	7	5	1	3	6
4	4	757	2	4	7	5	3	1	6
5	5	645	2	3	6	4	1	5	7
6	6	771	2	4	6	5	3	1	7
7	7	2	3	6	4	2	5	1	7
8	8	356	2	3	6	4	1	5	7
9	9	947	2	4	7	5	3	1	6
10	10	803	2	3	7	6	4	1	5

POWER BI

The screenshot shows the Microsoft Power BI Desktop interface. The main view displays a horizontal bar chart titled 'Total Investors by Avenue'. The chart shows the total number of investors for four investment avenues: Mutual fund, Fixed deposits, Equity, and Public provident fund. The x-axis represents the 'Total Investors' count, ranging from 0 to 300. The y-axis lists the investment avenues. The chart indicates that Mutual fund is the most popular avenue, followed by Fixed deposits, Equity, and Public provident fund. The status bar at the bottom shows 'Page 1 of 5' and the date '4/21/2026'.

Shows which investment avenue (Mutual Fund, Equity, FD, PPF) is most popular.





Decision-Making Based on Investment Dashboard Charts

Chart 1: Investors Distribution by Investment Avenue

Expected Insight:

Mutual Funds and Fixed Deposits are likely to dominate, followed by Equity and PPF.

Key Decisions:

- **If Mutual Funds rank first:**
 - Prioritize Mutual Fund offerings in financial portfolios
 - Launch targeted education campaigns to increase awareness
 - Introduce beginner-friendly plans for younger investors (ages 21–27)
- **If Fixed Deposits are highly preferred:**
 - Indicates a risk-averse audience → focus on capital-protected products
 - Offer competitive interest rates to retain customers
 - Introduce hybrid solutions (FD + Mutual Funds) for gradual risk exposure
- **If Equity adoption is low:**
 - Suggests low confidence in stock markets
 - Provide stock market awareness programs
 - Offer professionally managed equity portfolios

Chart 2: Investment Factors (Risk vs. Returns)

Expected Insight:

“Returns” is likely the dominant decision-making factor.

Key Decisions:

- **If Returns dominate:**
 - Emphasize high-return investment products
 - Showcase historical performance data
 - Provide comparison tools for expected returns
 - Focus advisory discussions on ROI projections
- **If Risk is a major factor:**
 - Introduce risk assessment tools before investment
 - Promote diversified portfolios

- Offer insurance-linked investment products
- Provide regular portfolio performance reports

Chart 3: Average Preference Scores by Investment Type

Expected Insight:

Different investment avenues will show varying preference rankings (1–7).

Key Decisions:

- **If Mutual Fund investors avoid Equity:**
 - Introduce Equity SIPs as a gradual entry point
 - Offer bundled investment packages (MF + Equity)
- **If Equity investors avoid Bonds:**
 - Focus on high-growth investment opportunities
 - Avoid overly conservative product recommendations
 - Provide advanced analytics and market insights
- **If PPF investors avoid other assets:**
 - Target them with long-term, low-risk products
 - Recommend tax-saving Mutual Funds (ELSS)
 - Highlight safety and tax benefits

Chart 4: Savings Objectives & Information Sources

Expected Insight:

Retirement planning is likely the top objective, with Financial Consultants as a key source.

Key Decisions:

- **If Retirement is the top goal:**
 - Develop long-term retirement investment plans
 - Offer automated portfolio rebalancing
 - Introduce corporate retirement programs
- **If Healthcare is a major objective:**
 - Create health-linked financial products
 - Encourage emergency fund strategies
 - Integrate investment with insurance solutions
- **If Financial Consultants dominate:**

- Invest in advisor training and development
- Provide incentive programs
- Equip advisors with comparison tools
- **If Media channels dominate:**
 - Increase investment in TV and print campaigns
 - Use financial influencers
 - Publish educational financial content

Chart 5: Wealth Creation KPI

Expected Insight:

A large percentage (≈70–80%) of investors focus on wealth creation.

Key Decisions:

- **If above 70%:**
 - Focus on high-return investment products
 - Develop structured wealth-building roadmaps
 - Offer tiered investment plans (Starter / Growth / Premium)
- **If below 50%:**
 - Emphasize capital protection and stability
 - Promote low-risk investment options
 - Provide simple entry-level financial products

Strategic Summary

Insight	Recommended Action
Mutual Funds dominate	Expand offerings and advisor expertise
High FD adoption	Introduce hybrid investment solutions
Returns > Risk	Focus marketing on ROI
Retirement is primary goal	Develop retirement-focused plans
Consultants are key source	Strengthen advisor networks
Young investors dominate	Build digital-first platforms
High return expectations	Educate investors on realistic returns